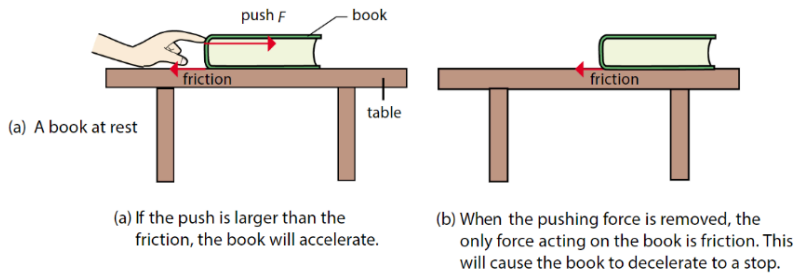


Newton's Second Law

if resultant force acting on an object is not 0, we say forces on the object are unbalanced.



Forces acting on book are balanced $\Rightarrow$ and it moves at constant velocity.

if applied force $F \Rightarrow$ is now increased $\Rightarrow$ forces that act on the book $\Rightarrow$ no longer balanced $\Rightarrow$ and book accelerates.

If applied force F is removed $\Rightarrow$ while book is still in motion $\rightarrow$ friction is the resultant force.

Resultant force $\Rightarrow$ causes book to decelerate and eventually stop .

Definition: Newton's second law states that when a resultant force acts on an object of a constant mass, the object will accelerate in the direction of the resultant force.

#PhysicsDefinitions

Product of mass and acceleration of object gives resultant force.

$$F = ma$$

F = resultant force (N)

m = mass of object (Kg)

a = acceleration of object (m/s^2)

#physicsFormulae

- resultant force F of an object $\Rightarrow$ produces an acceleration, a
 - Doubling the resultant force F of an object doubles its acceleration
 - with the same resultant Force F , doubling mass m halves the acceleration, a
-

Force $\Rightarrow$ is directly proportional to acceleration

Force $\Rightarrow$ is inversely proportional to mass.

- inverse proportion graph is a downward curve

